GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Comprehensive GATE Notes: Engineering Mathematics

Paper Code: EC • Section 1: Engineering Mathematics (13-15 Marks)

Subject Code: EC-MATH • **Target Exam:** GATE ECE • **Weightage:** 13-15 Marks

1. Linear Algebra: Matrix Operations & Spectral Theory

- **Matrix Inversion & Determinants:** For an n x n matrix A, det(A) = product of eigenvalues. A is invertible iff det(A) != 0. Key properties: det(AB) = det(A)det(B), det(kA) = k^n det(A), det(A^T) = det(A), det(A^-1) = 1/det(A).
- **Rank of a Matrix:** The rank rho(A) is the maximum number of linearly independent row/column vectors. For a linear system Ax = b: Unique solution iff rank(A) = rank([A|b]) = n. Infinitely many solutions iff rank(A) = rank([A|b]) < n (with n - rank free variables). No solution (inconsistent) iff rank([A|b]) > rank(A).
- **Eigenvalues & Cayley-Hamilton Theorem:** Roots of det(A - lambda*I) = 0. Trace(A) = sum of lambda_i, det(A) = product of lambda_i. Cayley-Hamilton states every square matrix satisfies its own characteristic equation: P(A) = 0. This enables fast computation of A^-1, A^4, and matrix polynomials.

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Characteristic Equation	**det(A - lambda*I) = 0**	lambda = Eigenvalues	Roots yield matrix spectral spectrum
Cayley-Hamilton Identity	**A^n + c_{n-1}A^{n-1} + ... + c_0 I = 0**	A = Matrix, I = Identity	Substitute A to find high powers or inverse
Orthogonal Matrix	**A^T * A = I, |det(A)| = 1**	A^T = Transpose	Eigenvalues lie on unit circle (|lambda| = 1)

GATE PYQ (2 Marks) - Cayley-Hamilton Application
Question: Consider the matrix A = [[2, 1], [1, 2]]. Find the eigenvalues and calculate A^4 using the Cayley-Hamilton theorem.

Step 1 (Char Eq): det(A - lambda*I) = (2-lambda)^2 - 1 = lambda^2 - 4*lambda + 3 = 0.
Step 2 (Eigenvalues): Factoring gives (lambda - 3)(lambda - 1) = 0 => **lambda_1 = 3, lambda_2 = 1**.
Step 3 (Cayley-Hamilton): A^2 - 4A + 3I = 0 => A^2 = 4A - 3I.
Step 4 (Higher Powers): A^3 = 4A^2 - 3A = 4(4A - 3I) - 3A = 13A - 12I.
Step 5: A^4 = 13A^2 - 12A = 13(4A - 3I) - 12A = 40A - 39I.
Step 6 (Evaluation): 40*[[2, 1], [1, 2]] - [[39, 0], [0, 39]] = **[[41, 40], [40, 41]]**.

2. Vector Calculus: Gradient, Divergence, Curl & Integral Theorems

- **Gradient (grad phi):** Vector perpendicular to level surface phi(x,y,z) = c in the direction of maximum rate of increase. Directional derivative in unit direction u is D_u phi = grad phi . u.
- **Divergence & Curl:** div F = grad . F (scalar, volume expansion rate); curl F = grad x F (vector, local rotational circulation). If curl F = 0, F is conservative/irrotational and F = grad phi.
- **Vector Integral Theorems:**

• *Green's Theorem:* oint_C (P dx + Q dy) = iint_R (dQ/dx - dP/dy) dA.
• *Stokes' Theorem:* oint_C F . dr = iint_S (curl F) . dS.
• *Gauss Divergence Theorem:* oiint_S F . dS = iiint_V (div F) dV.

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Gauss Divergence	**oiint_S F . dS = iiint_V (grad . F) dV**	S = Closed Surface, V = Volume	Converts closed surface flux into 3D volume integral
Stokes' Theorem	**oint_C F . dr = iint_S (grad x F) . dS**	C = Boundary Curve, S = Open Surface	Converts line circulation around boundary into surface curl
Irrotational Field	**curl F = 0 <=> F = grad phi**	phi = Scalar Potential	Work done along closed path is identically zero

GATE PYQ (2 Marks) - Gauss Divergence Application
Question: Evaluate the flux of vector field F = 2x i + 3y j + z k through the closed surface of the unit sphere x^2 + y^2 + z^2 = 1.

Step 1: By Gauss Divergence Theorem: Flux = iiint_V (grad . F) dV.
Step 2: div F = d(2x)/dx + d(3y)/dy + d(z)/dz = 2 + 3 + 1 = 6.
Step 3: Flux = iiint_V 6 dV = 6 * Volume(Sphere).
Step 4: Vol = (4/3)*pi*r^3 = (4/3)*pi*(1)^3 = 4*pi/3.
Step 5: Flux = 6 * (4*pi/3) = **8*pi** ≈ **25.133**.

3. Probability & Statistics: Bayes' Theorem & Distributions

- **Conditional Probability & Bayes' Theorem:** P(A|B) = P(A cap B)/P(B). Bayes' Rule: P(B_i|A) = [P(A|B_i)*P(B_i)] / sum_j [P(A|B_j)*P(B_j)]. Highly tested in digital communications bit error rates.
- **Continuous Random Variables:** PDF f_X(x) >= 0, int_{-inf}^{inf} f_X(x) dx = 1. Expectation E[X] = int x f_X(x) dx, Var(X) = E[X^2] - (E[X])^2.
- **Normal (Gaussian) Distribution:** N(mu, sigma^2). Q(x) = (1/sqrt(2pi)) int_x^inf e^{-u^2/2} du. Standardized Z = (X - mu)/sigma.

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Bayes' Theorem	**P(B_i|A) = P(A|B_i)P(B_i) / sum P(A|B_k)P(B_k)**	B_i = Partition, A = Observed Event	Updates prior probability with new evidence
Poisson Distribution	**P(X = k) = (lambda^k * e^{-lambda}) / k!**	lambda = Mean Rate	E[X] = lambda, Var(X) = lambda
Uniform Variance	**Var(X) = (b - a)^2 / 12**	a, b = Interval Bounds	For uniform RV on [a, b]; mean is (a+b)/2

GATE PYQ (1 Mark) - Bayes' Theorem in Binary Channel
Question: A binary symmetric channel has bit error probability p = 0.05. If 0 is transmitted with probability 0.6 and 1 with 0.4, find the probability that 0 was transmitted given that 0 was received.

Step 1: Events: T0, T1, R0, R1. Given P(T0) = 0.6, P(T1) = 0.4.
Step 2: Channel: P(R0|T0) = 1 - p = 0.95; P(R0|T1) = p = 0.05.
Step 3: Total prob P(R0) = 0.95*0.6 + 0.05*0.4 = 0.57 + 0.02 = 0.59.
Step 4: By Bayes' Rule: P(T0|R0) = [P(R0|T0)*P(T0)] / P(R0) = 0.57 / 0.59 = **0.9661 (96.61%)**.

Essential GATE Exam Pitfalls & Scoring Traps

- **Important:** Always check if det(A) = 0 before computing inverses or solving Cramer's rule.
- **Important:** In vector integration, verify if the boundary curve is oriented counterclockwise before applying Green's or Stokes' theorem.
- **Important:** For Poisson distributions, remember Mean = Variance = lambda.
- **Important:** In differential equations, Cauchy-Euler equations x^2 y'' + ax y' + by = 0 are solved by setting x = e^z.